

The Eye-tracking-based Adaptation for Personalized Game-based Learning within Intelligent Educational Environments

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Abstract. Today’s educational environments are taking advantage of various technologies that considerably improve interactivity and adaptability, and enable intelligent and personalized learning applications. The technology plays a pivotal role in game-based education, enabling personalized learning experiences. The development of eye-tracking technology in recent years has made its integration into educational environments a promising approach to enhance personalized learning, especially in the context of learning games. Various studies in the field of eye-tracking technology report on its ability to successfully collect real-time data on visual attention and gaze, as well as for pattern detection in eye movements, saccades, and gaze behavior, to identify topics or areas of interest and potential challenges faced by learners. Next, this information helps the adaptive game engine to tailor both the gameplay and related learning content that are integrated within the game. Thus, through adjustment of the challenges to align with both the current cognitive and emotional state of a particular learner, the adaptive educational game is expected to foster their sustained engagement in achieving improved learning motivation and gaming experience. This paper explores a broader approach to eye-tracking sensor usage for educational purposes and briefly reviews its potential applications. It especially outlines the integration of this technology and its applications in game-based learning to facilitate adaptive learning strategies within intelligent educational environments.